STAT 4355
Linear Regression

Lecture Notes

Yoolkyu Park

Contents

1 Simple Linear Regression and Least Squares **1**
- 1.1 The regression model . . . 1
- 1.2 Mean, variance, and covariance . . . 1
- 1.3 Least squares estimation . . . 2
- 1.4 Corrected sums and the fitted line . . . 4

2 Sampling Properties and Variance Estimation **6**
- 2.1 Estimators are random variables . . . 6
- 2.2 The slope as a linear combination of the observations . . . 6
- 2.3 The intercept and a centered model . . . 8
- 2.4 Residuals and estimation of σ^2 . . . 9

3 Hypothesis Testing in Simple Regression **11**
- 3.1 Null distributions and rejection regions . . . 11
- 3.2 p-values . . . 12
- 3.3 The normal distribution of the slope estimator . . . 12
- 3.4 Estimating the variance and the t distribution . . . 14

4 Confidence Intervals, Prediction, and Analysis of Variance **16**
- 4.1 From point estimation to interval estimation . . . 16
- 4.2 Confidence intervals for the regression parameters . . . 17
- 4.3 Confidence interval on the mean response . . . 18
- 4.4 Prediction of a new observation . . . 19
- 4.5 Analysis of variance . . . 20
- 4.6 Regression through the origin . . . 22

5 Maximum Likelihood Estimation **24**
- 5.1 Maximum likelihood for a normal sample . . . 24
- 5.2 Likelihood and least squares . . . 25

6 Matrix Formulation and Geometry of Linear Regression **27**
- 6.1 Multiple linear regression in matrix form . . . 27
- 6.2 Least squares and orthogonal projection . . . 28
- 6.3 Expectation and covariance of random vectors . . . 30
- 6.4 Properties of the least squares estimator . . . 31

Chapter 1

Simple Linear Regression and Least Squares

1.1 The regression model

Regression analysis describes a relationship between a response and a predictor. A deterministic straight-line relationship has the form

$$y = \beta_0 + \beta_1 x.$$

To allow variation around the line, we write

$$Y_i = \beta_0 + \beta_1 x_i + \varepsilon_i, \qquad i = 1, \ldots, n. \tag{1.1}$$

The predictor is also called an independent variable, covariate, or regressor. The values x_i and the responses y_i are observable. The coefficients β_0 and β_1 are unknown, fixed parameters. The intercept is β_0, and the slope is β_1.

We treat the predictor values $x_1, \ldots, x_n$ as fixed. Before observing the data, Y_i is a random variable; y_i denotes its observed value. The error ε_i represents the deviation of Y_i from the regression line.

Definition 1.1 (Error assumptions)**.** The simple linear regression model with uncorrelated, equal-variance errors assumes

$$\mathbb{E}(\varepsilon_i) = 0, \qquad \mathrm{Var}(\varepsilon_i) = \sigma^2 > 0, \qquad \mathrm{Cov}(\varepsilon_i, \varepsilon_j) = 0 \quad (i \neq j).$$

When independence or normality is needed, it will be stated explicitly.

Note 1.2. For random predictors, the corresponding assumptions concern the errors conditional on the predictors. In particular, $\mathbb{E}(\varepsilon_i \mid x_1, \ldots, x_n) = 0$ gives the conditional mean model. With fixed predictors, no probability distribution for the x_i is required.

1.2 Mean, variance, and covariance

Since $\beta_0 + \beta_1 x_i$ is constant,

$$\begin{aligned} \mathbb{E}(Y_i) &= \mathbb{E}(\beta_0 + \beta_1 x_i + \varepsilon_i) \\ &= \beta_0 + \beta_1 x_i + \mathbb{E}(\varepsilon_i) = \beta_0 + \beta_1 x_i, \\ \mathrm{Var}(Y_i) &= \mathrm{Var}(\beta_0 + \beta_1 x_i + \varepsilon_i) = \mathrm{Var}(\varepsilon_i) = \sigma^2. \end{aligned}$$

Thus the regression line describes the mean response. Equal error variances mean that the spread around this line does not change with the predictor level. This is called homoscedasticity. If the spread changes with x, the errors are heteroscedastic.

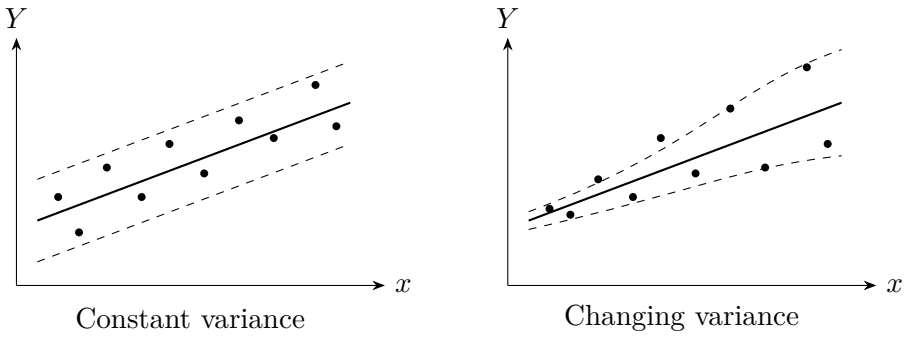


Figure 1.1: The same linear mean structure can occur with different error variances. The dashed curves indicate spread, not confidence limits.

Recall that for random variables U, V with finite second moments,

$$\mathrm{Var}(U) = \mathbb{E}\big[(U - \mathbb{E}U)^2\big], \qquad \mathrm{Cov}(U, V) = \mathbb{E}\big[(U - \mathbb{E}U)(V - \mathbb{E}V)\big].$$

Adding a constant shifts the center of a distribution without changing its variance. Likewise,

$$\mathrm{Cov}(Y_i, Y_j) = \mathrm{Cov}(\varepsilon_i, \varepsilon_j) = 0 \quad (i \neq j).$$

Uncorrelated errors have no linear association across observations. Independence is a stronger condition; zero covariance alone does not imply independence.

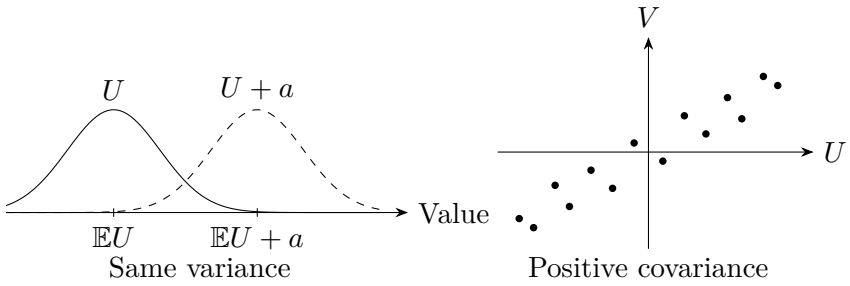


Figure 1.2: Left: translating a distribution changes its mean but not its variance. Right: a positive covariance pattern for centered variables U, V. The points are schematic; this pattern differs from the uncorrelated errors in the regression model.

1.3 Least squares estimation

We define the least squares criterion by

$$S(b_0, b_1) = \sum_{i=1}^{n} (y_i - b_0 - b_1 x_i)^2. \tag{1.2}$$

By minimizing S, we obtain the least squares estimates $\widehat{\beta}_0$ and $\widehat{\beta}_1$. Define

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i, \qquad \bar{y} = \frac{1}{n}\sum_{i=1}^{n} y_i,$$
$$S_{xx} = \sum_{i=1}^{n}(x_i - \bar{x})^2, \qquad S_{xy} = \sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y}).$$

Assume $S_{xx} > 0$, so that the predictor values are not all equal.

Proposition 1.3 (Normal equations and estimates)**.** *The least squares estimates satisfy*

$$n\widehat{\beta}_0 + \widehat{\beta}_1 \sum_{i=1}^{n} x_i = \sum_{i=1}^{n} y_i, \qquad \widehat{\beta}_0 \sum_{i=1}^{n} x_i + \widehat{\beta}_1 \sum_{i=1}^{n} x_i^2 = \sum_{i=1}^{n} x_i y_i. \tag{1.3}$$

Consequently,

$$\widehat{\beta}_1 = \frac{S_{xy}}{S_{xx}}, \qquad \widehat{\beta}_0 = \bar{y} - \widehat{\beta}_1 \bar{x}. \tag{1.4}$$

Proof. The first derivatives are

$$\frac{\partial S}{\partial b_0} = -2\sum_{i=1}^{n}(y_i - b_0 - b_1 x_i),$$
$$\frac{\partial S}{\partial b_1} = -2\sum_{i=1}^{n} x_i(y_i - b_0 - b_1 x_i).$$

Setting both derivatives equal to zero gives (1.3). The first equation yields

$$\widehat{\beta}_0 = \frac{\sum_i y_i - \widehat{\beta}_1 \sum_i x_i}{n} = \bar{y} - \widehat{\beta}_1 \bar{x}.$$

Substituting this expression into the second equation gives

$$\widehat{\beta}_1 \left(\sum_i x_i^2 - n\bar{x}^2 \right) = \sum_i x_i y_i - n\bar{x}\bar{y}.$$

The expressions in parentheses and on the right are S_{xx} and S_{xy}, respectively. Since $S_{xx} > 0$, division gives (1.4).

Exercise 1.1 (Checking the minimum)**.** Verify that the solution of the normal equations is the unique global minimizer of S when $S_{xx} > 0$. What happens when $S_{xx} = 0$?

Solution. The Hessian of S is

$$\nabla^2 S = 2\begin{pmatrix} n & \sum_i x_i \\ \sum_i x_i & \sum_i x_i^2 \end{pmatrix}.$$

For $(u, v) \neq (0, 0)$,

$$\frac{1}{2}\begin{pmatrix} u & v \end{pmatrix} \nabla^2 S \begin{pmatrix} u \\ v \end{pmatrix} = \sum_i (u + v x_i)^2 = n(u + v\bar{x})^2 + v^2 S_{xx} > 0.$$

Hence S is strictly convex and its stationary point is the unique global minimum. Equivalently, the leading principal minors are $2n > 0$ and $4nS_{xx} > 0$.

If $S_{xx} = 0$, all $x_i = c$ for some constant c. Only the combination $b_0 + cb_1$ is identifiable. The minimum occurs whenever $b_0 + cb_1 = \bar{y}$, so there is no unique pair of coefficients.

1.4 Corrected sums and the fitted line

The corrected sum of squares has the equivalent forms

$$\begin{aligned} S_{xx} &= \sum_{i=1}^{n}(x_i - \bar{x})^2 \\ &= \sum_i x_i^2 - 2\bar{x}\sum_i x_i + n\bar{x}^2 \\ &= \sum_i x_i^2 - n\bar{x}^2 = \sum_i x_i^2 - \frac{(\sum_i x_i)^2}{n}. \end{aligned}$$

Similarly, because $\sum_i (x_i - \bar{x}) = 0$,

$$\begin{aligned} S_{xy} &= \sum_i (x_i - \bar{x})(y_i - \bar{y}) \\ &= \sum_i x_i y_i - \bar{y}\sum_i x_i - \bar{x}\sum_i y_i + n\bar{x}\bar{y} \\ &= \sum_i x_i y_i - n\bar{x}\bar{y} = \sum_i x_i y_i - \frac{(\sum_i x_i)(\sum_i y_i)}{n}. \end{aligned}$$

Thus the slope may be computed as

$$\widehat{\beta}_1 = \frac{\sum_i x_i y_i - \dfrac{(\sum_i x_i)(\sum_i y_i)}{n}}{\sum_i x_i^2 - \dfrac{(\sum_i x_i)^2}{n}}.$$

For $n > 1$, the sample variance and sample covariance are

$$s_x^2 = \frac{S_{xx}}{n-1}, \qquad s_{xy} = \frac{S_{xy}}{n-1}.$$

Therefore $\widehat{\beta}_1 = s_{xy}/s_x^2$. This scalar ratio applies to simple linear regression with an intercept.

Definition 1.4 (Fitted values and residuals)**.** The fitted simple linear regression line is

$$\widehat{y}(x) = \widehat{\beta}_0 + \widehat{\beta}_1 x.$$

At an observed predictor value x_i, the fitted value is $\widehat{y}_i = \widehat{\beta}_0 + \widehat{\beta}_1 x_i$. The residual is the difference between the observed and fitted responses:

$$e_i = y_i - \widehat{y}_i = y_i - (\widehat{\beta}_0 + \widehat{\beta}_1 x_i).$$

Exercise 1.2 (Residual identities)**.** Show that the fitted line passes through $(\bar{x}, \bar{y})$, and that

$$\sum_i e_i = 0, \qquad \sum_i x_i e_i = 0, \qquad \sum_i \widehat{y}_i e_i = 0.$$

Solution. From $\widehat{\beta}_0 = \bar{y} - \widehat{\beta}_1 \bar{x}$, we have $\widehat{y}(\bar{x}) = \bar{y}$. The two normal equations give

$$\sum_i e_i = \sum_i (y_i - \widehat{\beta}_0 - \widehat{\beta}_1 x_i) = 0, \qquad \sum_i x_i e_i = 0.$$

Consequently,

$$\sum_i \widehat{y}_i e_i = \widehat{\beta}_0 \sum_i e_i + \widehat{\beta}_1 \sum_i x_i e_i = 0.$$

These identities are algebraic properties of least squares with an intercept; they do not require normal errors.

Note 1.5. An error ε_i is the deviation from the unknown population regression line. A residual e_i is the deviation from the fitted line. They are different quantities, even though residuals help assess the error assumptions.

Chapter 2

Sampling Properties and Estimation of the Error Variance

2.1 Estimators are random variables

Recall the simple regression model (1.1) with fixed design points and the error assumptions in Definition 1.1. Assume additionally that the errors are independent and $S_{xx} > 0$. For random responses, write

$$\bar{Y} = \frac{1}{n}\sum_{i=1}^{n} Y_i, \qquad S_{xy} = \sum_{i=1}^{n}(x_i - \bar{x})(Y_i - \bar{Y}).$$

The parameters β_0 and β_1 are unknown constants. Before collecting data, $\widehat{\beta}_0$ and $\widehat{\beta}_1$ are random variables. After collecting data, their observed values are estimates. A second sample of responses at the same design points generally produces different estimates. We wish to find

$$\mathbb{E}(\widehat{\beta}_0), \quad \mathrm{Var}(\widehat{\beta}_0), \quad \mathbb{E}(\widehat{\beta}_1), \quad \mathrm{Var}(\widehat{\beta}_1).$$

Normality is not needed for these expectations and variances.

2.2 The slope as a linear combination of the observations

The least squares estimators are

$$\widehat{\beta}_1 = \frac{S_{xy}}{S_{xx}}, \qquad \widehat{\beta}_0 = \bar{Y} - \bar{x}\widehat{\beta}_1.$$

Put

$$c_i = \frac{x_i - \bar{x}}{S_{xx}}.$$

These are constants because the design points are fixed. They satisfy

$$\sum_{i=1}^{n} c_i = \frac{\sum_i x_i - n\bar{x}}{S_{xx}} = 0,$$
$$\sum_{i=1}^{n} c_i x_i = \frac{\sum_i (x_i - \bar{x})\{(x_i - \bar{x}) + \bar{x}\}}{S_{xx}} = \frac{S_{xx} + 0}{S_{xx}} = 1,$$
$$\sum_{i=1}^{n} c_i^2 = \frac{1}{S_{xx}}.$$

Consequently,

$$\widehat{\beta}_1 = \sum_{i=1}^{n} c_i (Y_i - \bar{Y}) = \sum_{i=1}^{n} c_i Y_i - \bar{Y} \sum_{i=1}^{n} c_i = \sum_{i=1}^{n} c_i Y_i.$$

Proposition 2.1. *The least squares slope estimator is unbiased, and its variance is*

$$\mathbb{E}(\widehat{\beta}_1) = \beta_1, \qquad \operatorname{Var}(\widehat{\beta}_1) = \frac{\sigma^2}{S_{xx}}.$$

Proof. By linearity of expectation,

$$\begin{aligned}\mathbb{E}(\widehat{\beta}_1) &= \mathbb{E}\left(\sum_{i=1}^{n} c_i Y_i\right) = \sum_{i=1}^{n} c_i \mathbb{E}(Y_i) \\ &= \sum_{i=1}^{n} c_i (\beta_0 + \beta_1 x_i) = \beta_0 \sum_{i=1}^{n} c_i + \beta_1 \sum_{i=1}^{n} c_i x_i = \beta_1.\end{aligned}$$

For $i \neq j$,

$$\operatorname{Cov}(Y_i, Y_j) = \operatorname{Cov}(\varepsilon_i, \varepsilon_j) = 0.$$

Thus

$$\operatorname{Var}(\widehat{\beta}_1) = \sum_{i=1}^{n} c_i^2 \operatorname{Var}(Y_i) = \sigma^2 \sum_{i=1}^{n} c_i^2 = \frac{\sigma^2}{S_{xx}}.$$

Example 2.2 (The variance rule for two observations)**.** For two independent observations,

$$\begin{aligned}\operatorname{Var}(c_1 Y_1 + c_2 Y_2) &= c_1^2 \operatorname{Var}(Y_1) + c_2^2 \operatorname{Var}(Y_2) + 2 \operatorname{Cov}(c_1 Y_1, c_2 Y_2) \\ &= c_1^2 \sigma^2 + c_2^2 \sigma^2 + 2 c_1 c_2 \cdot 0 = (c_1^2 + c_2^2)\sigma^2.\end{aligned}$$

The same calculation gives the variance rule used for n observations.

The spread of the design points controls the precision of the slope estimate: larger S_{xx} gives smaller variance.

Exercise 2.1. Append a design point x_{n+1} to the existing n design points. Show that

$$S_{xx,\,n+1} = S_{xx,\,n} + \frac{n}{n+1}(x_{n+1} - \bar{x}_n)^2.$$

What happens to the slope variance when the error variance stays the same?

Solution. Write $d = x_{n+1} - \bar{x}_n$. The new mean is $\bar{x}_{n+1} = \bar{x}_n + d/(n+1)$, so

$$\begin{aligned} S_{xx,\,n+1} &= \sum_{i=1}^{n} \left(x_i - \bar{x}_n - \frac{d}{n+1} \right)^2 + \left(d - \frac{d}{n+1} \right)^2 \\ &= S_{xx,\,n} - \frac{2d}{n+1} \sum_{i=1}^{n} (x_i - \bar{x}_n) + \frac{nd^2}{(n+1)^2} + \frac{n^2 d^2}{(n+1)^2} \\ &= S_{xx,\,n} + \frac{n}{n+1} d^2. \end{aligned}$$

Thus S_{xx} cannot decrease, and σ^2/S_{xx} cannot increase. The slope variance decreases strictly when $x_{n+1} \neq \bar{x}_n$ and stays unchanged when the new point is the old mean.

2.3 The intercept and a centered model

Both least squares estimators are linear combinations of the observations. In particular,

$$\widehat{\beta}_0 = \sum_{i=1}^{n} \left(\frac{1}{n} - \bar{x} c_i \right) Y_i.$$

Exercise 2.2. Show that

$$\mathbb{E}(\widehat{\beta}_0) = \beta_0, \qquad \operatorname{Var}(\widehat{\beta}_0) = \sigma^2 \left(\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}} \right).$$

Identify the covariance term in the variance calculation.

Solution. First,

$$\mathbb{E}(\bar{Y}) = \beta_0 + \beta_1 \bar{x},$$

so

$$\mathbb{E}(\widehat{\beta}_0) = \mathbb{E}(\bar{Y}) - \bar{x}\mathbb{E}(\widehat{\beta}_1) = \beta_0 + \beta_1 \bar{x} - \bar{x}\beta_1 = \beta_0.$$

Independence gives $\operatorname{Var}(\bar{Y}) = \sigma^2/n$, and

$$\operatorname{Cov}(\bar{Y}, \widehat{\beta}_1) = \operatorname{Cov}\left(\frac{1}{n} \sum_i Y_i, \sum_j c_j Y_j \right) = \frac{\sigma^2}{n} \sum_i c_i = 0.$$

The three terms in the intercept variance are therefore

$$\begin{aligned} \operatorname{Var}(\widehat{\beta}_0) &= \operatorname{Var}(\bar{Y}) + \bar{x}^2 \operatorname{Var}(\widehat{\beta}_1) - 2\bar{x} \operatorname{Cov}(\bar{Y}, \widehat{\beta}_1) \\ &= \frac{\sigma^2}{n} + \bar{x}^2 \frac{\sigma^2}{S_{xx}} + 0 = \sigma^2 \left(\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}} \right). \end{aligned}$$

An alternative form of the model shifts the center to $\bar{x}$:

$$\begin{aligned} Y_i &= \beta_0 + \beta_1 (x_i - \bar{x}) + \beta_1 \bar{x} + \varepsilon_i \\ &= (\beta_0 + \beta_1 \bar{x}) + \beta_1 (x_i - \bar{x}) + \varepsilon_i \\ &= \beta_0^* + \beta_1 (x_i - \bar{x}) + \varepsilon_i, \qquad \beta_0^* = \beta_0 + \beta_1 \bar{x}. \end{aligned}$$

The centered intercept is the mean response at $x = \bar{x}$, which is often easier to interpret than the mean response at $x = 0$. Its estimator is

$$\widehat{\beta}_0^* = \widehat{\beta}_0 + \bar{x}\widehat{\beta}_1 = \bar{Y}, \qquad \operatorname{Var}(\widehat{\beta}_0^*) = \frac{\sigma^2}{n}.$$

2.4 Residuals and estimation of σ^2

Write $\widehat{Y}_i = \widehat{\beta}_0 + \widehat{\beta}_1 x_i$ and $e_i = Y_i - \widehat{Y}_i$. Recall from Exercise 1.2 that

$$\sum_i e_i = 0, \qquad \sum_i x_i e_i = 0, \qquad \sum_i \widehat{Y}_i e_i = 0, \qquad \widehat{\beta}_0 + \widehat{\beta}_1 \bar{x} = \bar{Y}.$$

These are algebraic properties of the fit with an intercept.

For independent, identically distributed observations W_i with variance σ^2, the sample variance

$$\frac{1}{n-1}\sum_{i=1}^{n}(W_i - \bar{W})^2$$

is unbiased for σ^2. In regression the responses have different means, so we use residuals to mimic the behavior of the errors. Define the sum of squared errors and the mean square error by

$$\mathrm{SSE} = \sum_{i=1}^{n} e_i^2, \qquad s^2 = \widehat{\sigma}^2 = \frac{\mathrm{SSE}}{n-2} = \mathrm{MSE}, \qquad n > 2.$$

With $S_{yy} = \sum_i (Y_i - \bar{Y})^2$, a useful computational form is

$$\mathrm{SSE} = S_{yy} - \frac{S_{xy}^2}{S_{xx}}, \qquad S_{yy} = \sum_{i=1}^{n} Y_i^2 - \frac{\left(\sum_{i=1}^{n} Y_i\right)^2}{n}.$$

Indeed, $e_i = (Y_i - \bar{Y}) - \widehat{\beta}_1 (x_i - \bar{x})$, so

$$\sum_i e_i^2 = S_{yy} - 2\widehat{\beta}_1 S_{xy} + \widehat{\beta}_1^2 S_{xx} = S_{yy} - \frac{S_{xy}^2}{S_{xx}}.$$

Exercise 2.3. Prove that $\mathbb{E}(\widehat{\sigma}^2) = \sigma^2$ without assuming normal errors. Explain the denominator $n-2$.

Solution. Put $\bar{\varepsilon} = n^{-1}\sum_i \varepsilon_i$ and

$$D = \widehat{\beta}_1 - \beta_1 = \frac{\sum_i (x_i - \bar{x})\varepsilon_i}{S_{xx}}.$$

Then

$$e_i = \varepsilon_i - \bar{\varepsilon} - D(x_i - \bar{x}).$$

Using $\sum_i (x_i - \bar{x}) = 0$ and $\sum_i (x_i - \bar{x})\varepsilon_i = D S_{xx}$ gives

$$\begin{aligned} \mathrm{SSE} &= \sum_i (\varepsilon_i - \bar{\varepsilon})^2 - 2D\sum_i (x_i - \bar{x})(\varepsilon_i - \bar{\varepsilon}) + D^2 S_{xx} \\ &= \sum_i \varepsilon_i^2 - n\bar{\varepsilon}^2 - D^2 S_{xx}. \end{aligned}$$

Now $\mathbb{E}(\bar{\varepsilon}) = \mathbb{E}(D) = 0$, and

$$\mathbb{E}(\bar{\varepsilon}^2) = \frac{\sigma^2}{n}, \qquad \mathbb{E}(D^2) = \frac{\sigma^2}{S_{xx}}.$$

Therefore

$$\mathbb{E}(\mathrm{SSE}) = n\sigma^2 - n\frac{\sigma^2}{n} - S_{xx}\frac{\sigma^2}{S_{xx}} = (n-2)\sigma^2,$$

and division by $n-2$ proves the result. Estimation of the intercept and slope removes two independent fitted directions; the residuals obey the two independent constraints $\sum_i e_i = 0$ and $\sum_i (x_i - \bar{x}) e_i = 0$. The residual degrees of freedom are $n-2$.

Exercise 2.4. Does $\mathbb{E}(s^2) = \sigma^2$ imply $\mathbb{E}(s) = \sigma$, where $s = \sqrt{s^2}$?

Solution. No. The square root is concave, so

$$\mathbb{E}(s) = \mathbb{E}(\sqrt{s^2}) \leq \sqrt{\mathbb{E}(s^2)} = \sigma.$$

The inequality is strict when s^2 is not almost surely constant, as in the normal error model with $n > 2$ and $\sigma^2 > 0$. Thus s^2 is unbiased for the variance, whereas s generally underestimates the standard deviation on average.

Chapter 3

Hypothesis Testing in Simple Regression

3.1 Null distributions and rejection regions

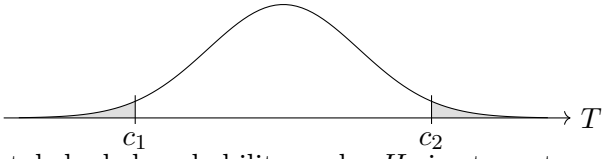
A hypothesis test asks whether the observed data are unusually extreme under a proposed null hypothesis H_0. If they are, we reject H_0; otherwise, we do not reject it. To judge whether an observation is rare, we choose a test statistic, a random variable

$$T = T(X_1, \ldots, X_n),$$

and obtain its distribution under H_0. For a two-sided test, choose cutoffs c_1 and c_2 such that

$$\mathbb{P}_{H_0}(T < c_1 \text{ or } T > c_2) \leq \alpha.$$

For example, $\alpha = 0.05$ bounds the probability of rejecting a true null hypothesis by 0.05.



Total shaded probability under H_0 is at most α.

Example 3.1 (Twenty Bernoulli observations)**.** Suppose $X_1, \ldots, X_{20}$ are independent and identically distributed Bernoulli(p) random variables. Test

$$H_0 : p = \frac{1}{2}, \qquad H_1 : p \neq \frac{1}{2}.$$

Under H_0, $K = \sum_{i=1}^{20} X_i$ has the binomial distribution with parameters 20 and 1/2. A two-sided rejection rule is $K < 6$ or $K > 14$.

Exercise 3.1. Compute the null probability of this rejection region and determine whether it gives a test at level 0.05.

Solution. Symmetry about 10 gives

$$\begin{aligned}\mathbb{P}_{H_0}(K < 6 \text{ or } K > 14) &= 2\sum_{k=0}^{5}\binom{20}{k}\left(\frac{1}{2}\right)^{20} \\ &= \frac{2(1+20+190+1140+4845+15504)}{1048576} \\ &= \frac{43400}{1048576} \approx 0.04139 < 0.05.\end{aligned}$$

Thus the test has level at most 0.05. The strict inequalities correspond to $K \leq 5$ or $K \geq 15$.

For a sample mean from independent $N(\mu, \sigma^2)$ observations with known σ^2, the null hypothesis $H_0 : \mu = \mu_0$ gives

$$Z = \frac{\bar{X} - \mu_0}{\mathrm{SD}(\bar{X})} = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}} \sim N(0,1), \qquad \mathrm{Var}(\bar{X}) = \mathrm{Var}\left(\frac{1}{n}\sum_i X_i\right) = \frac{\sigma^2}{n}.$$

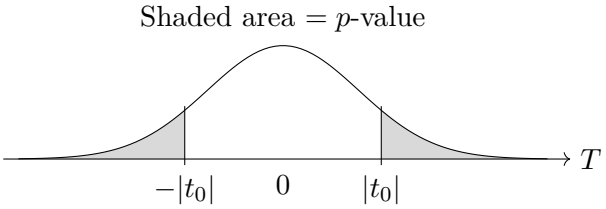
For a two-sided test at level 0.05, reject when $|Z| > 1.96$. For independent, identically distributed observations with finite, positive variance, the central limit theorem gives an approximate standard normal distribution after this standardization as n grows. Replacing the population variance by a sample estimate also requires justification; it does not generally give an exact normal distribution in a finite sample.

3.2 p-values

Under H_0, the probability that a new test statistic is equally or more extreme than the observed statistic t_0 is the p-value. Extremeness is determined by the alternative hypothesis. For a two-sided test with a symmetric null distribution centered at zero,

$$p\text{-value} = \mathbb{P}_{H_0}(|T| \geq |t_0|).$$

The two tail areas together form the p-value. Reject at significance level α when the p-value is at most α.

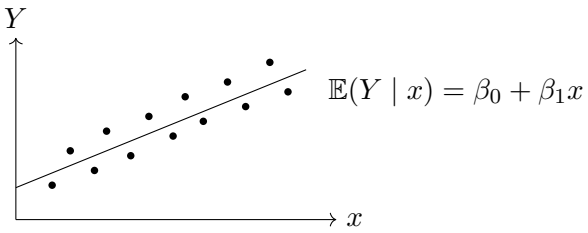


3.3 The normal distribution of the slope estimator

In simple regression, a basic question is

$$H_0 : \beta_1 = 0, \qquad H_1 : \beta_1 \neq 0.$$

Within the assumed linear mean model, $\beta_1 = 0$ means that the mean response does not change with x. A nonzero slope represents a linear change in the mean response.



For an exact finite-sample test we now impose the additional assumption

$$\varepsilon_i \text{ independent}, \qquad \varepsilon_i \sim N(0, \sigma^2).$$

The responses are independent normal random variables with means $\beta_0 + \beta_1 x_i$ and common variance σ^2; their means need not be identical.

Lemma 3.2. *If $Y_1, \dots, Y_n$ are independent normal random variables, with means μ_i and variances v_i, then a linear combination satisfies*

$$\sum_{i=1}^{n} a_i Y_i \sim N\left(\sum_{i=1}^{n} a_i \mu_i,\ \sum_{i=1}^{n} a_i^2 v_i\right)$$

whenever the resulting variance is positive.

Exercise 3.2. Prove the lemma using moment generating functions.

Solution. The moment generating function of $N(\mu_i, v_i)$ is $\exp(\mu_i t + v_i t^2/2)$. Independence gives

$$\begin{aligned}
\mathbb{E}\exp\left(t \sum_i a_i Y_i\right) &= \prod_i \mathbb{E}\exp(t a_i Y_i) \\
&= \prod_i \exp\left(t a_i \mu_i + \frac{t^2 a_i^2 v_i}{2}\right) \\
&= \exp\left(t \sum_i a_i \mu_i + \frac{t^2}{2} \sum_i a_i^2 v_i\right).
\end{aligned}$$

This is the moment generating function of the claimed normal distribution, which determines its distribution.

Since $\widehat{\beta}_1 = \sum_i c_i Y_i$, the lemma and the previously computed moments give

$$\widehat{\beta}_1 \sim N\left(\beta_1, \frac{\sigma^2}{S_{xx}}\right), \qquad \frac{\widehat{\beta}_1 - \beta_1}{\sqrt{\sigma^2/S_{xx}}} \sim N(0,1).$$

If σ^2 is known, the null statistic

$$Z_0 = \frac{\widehat{\beta}_1}{\sqrt{\sigma^2/S_{xx}}}$$

has a distribution under H_0 that does not depend on unknown parameters.

Example 3.3. Suppose the observed slope estimate is $\widehat{\beta}_1 = 2$ and its known standard deviation is $\sqrt{\sigma^2/S_{xx}} = 2$. The distance from the null value 0 is one standard deviation:

$$z_0 = \frac{2 - 0}{2} = 1.$$

For the two-sided alternative, the p-value is

$$\mathbb{P}(|Z| \geq 1) = 2\{1 - \Phi(1)\} \approx 0.3173,$$

where Φ is the standard normal distribution function. We do not reject H_0 at level 0.05.

3.4 Estimating the variance and the t distribution

Usually σ^2 is unknown. For $n > 2$, we replace it by $s^2 = \mathrm{SSE}/(n-2)$. Estimating the denominator introduces more variability, so the resulting exact distribution is no longer standard normal.

Recall that if $Z \sim N(0,1)$ and $U \sim \chi^2_\nu$ are independent, then

$$\frac{Z}{\sqrt{U/\nu}} \sim t_\nu.$$

Here χ^2_ν is the distribution of the sum of squares of ν independent standard normal random variables.

Exercise 3.3. Under the normal error model, with $n > 2$ and $S_{xx} > 0$, prove that

$$U = \frac{\mathrm{SSE}}{\sigma^2} \sim \chi^2_{n-2},$$

and that U is independent of

$$Z = \frac{\widehat{\beta}_1 - \beta_1}{\sigma/\sqrt{S_{xx}}}.$$

Use these facts to obtain the exact distribution after estimating σ^2.

Solution. In $\mathbb{R}^n$, let

$$u = \frac{1}{\sqrt{n}}(1, \ldots, 1), \qquad v = \frac{1}{\sqrt{S_{xx}}}(x_1 - \bar{x}, \ldots, x_n - \bar{x}).$$

These two vectors have length one and are orthogonal because $\sum_i (x_i - \bar{x}) = 0$. Complete them to an orthonormal basis $u, v, w_3, \ldots, w_n$.

Let $\eta = (\varepsilon_1/\sigma, \ldots, \varepsilon_n/\sigma)$. Its density is

$$(2\pi)^{-n/2} \exp\left(-\frac{\|\eta\|^2}{2}\right).$$

An orthogonal change of coordinates preserves squared length and has Jacobian of absolute value one. Therefore its coordinates

$$A = u^{\mathsf{T}}\eta, \qquad B = v^{\mathsf{T}}\eta, \qquad C_j = w_j^{\mathsf{T}}\eta \quad (j = 3, \ldots, n)$$

have joint density

$$(2\pi)^{-n/2}\exp\left[-\frac{1}{2}\left(A^2+B^2+\sum_{j=3}^{n}C_j^2\right)\right].$$

The density factors into n standard normal densities, so these coordinates are independent $N(0,1)$ variables.

Now

$$B=\frac{\sum_i(x_i-\bar{x})\varepsilon_i}{\sigma\sqrt{S_{xx}}}=\frac{\widehat{\beta}_1-\beta_1}{\sigma/\sqrt{S_{xx}}}=Z.$$

The standardized residual vector is obtained by subtracting the two fitted coordinates:

$$\frac{e}{\sigma}=\eta-Au-Bv=\sum_{j=3}^{n}C_jw_j,\qquad e=(e_1,\ldots,e_n).$$

Consequently,

$$\frac{\mathrm{SSE}}{\sigma^2}=\left\|\frac{e}{\sigma}\right\|^2=\sum_{j=3}^{n}C_j^2\sim\chi^2_{n-2}.$$

The fitted estimators depend only on the first two coordinates, since

$$\bar{Y}=\beta_0+\beta_1\bar{x}+\frac{\sigma A}{\sqrt{n}},\qquad\widehat{\beta}_1=\beta_1+\frac{\sigma B}{\sqrt{S_{xx}}},\qquad\widehat{\beta}_0=\bar{Y}-\bar{x}\widehat{\beta}_1.$$

Since A and B are jointly independent of all the C_j, the pair $(\bar{Y},\widehat{\beta}_1)$ is independent of this sum of squares. Consequently, the full coefficient vector and every fitted mean are independent of U. In particular, $Z=B$ is independent of U. Finally,

$$\frac{Z}{\sqrt{U/(n-2)}}=\frac{\widehat{\beta}_1-\beta_1}{s/\sqrt{S_{xx}}}\sim t_{n-2}.$$

Thus the estimated standard error of the slope and the null test statistic are

$$\mathrm{SE}(\widehat{\beta}_1)=\frac{s}{\sqrt{S_{xx}}},\qquad T_0=\frac{\widehat{\beta}_1}{\mathrm{SE}(\widehat{\beta}_1)}\sim t_{n-2}\quad\text{under } H_0:\beta_1=0.$$

For $H_1:\beta_1\neq 0$, reject when

$$|T_0|>t_{n-2,1-\alpha/2},$$

where $t_{\nu,q}$ denotes the q quantile of t_ν. If the observed statistic is t_0 and $F_{t_{n-2}}$ is its null distribution function, then

$$p\text{-value}=2\{1-F_{t_{n-2}}(|t_0|)\}.$$

Chapter 4

Confidence Intervals, Prediction, and Analysis of Variance

4.1 From point estimation to interval estimation

Consider the simple linear regression model

$$Y_i = \beta_0 + \beta_1 x_i + \varepsilon_i, \qquad i = 1, \ldots, n,$$

where the x_i are fixed and the errors are independent, with $\mathbb{E}(\varepsilon_i) = 0$ and $\mathrm{Var}(\varepsilon_i) = \sigma^2 > 0$. Assume $n > 2$ and

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i, \qquad S_{xx} = \sum_{i=1}^{n}(x_i - \bar{x})^2 > 0.$$

Write

$$\widehat{Y}_i = \widehat{\beta}_0 + \widehat{\beta}_1 x_i, \qquad e_i = Y_i - \widehat{Y}_i, \qquad s^2 = \widehat{\sigma}^2 = \mathrm{MSE} = \frac{\sum_{i=1}^{n} e_i^2}{n-2}.$$

Normality is needed for the exact normal, t, and chi-square distribution statements below. For those statements assume $\varepsilon_i \sim N(0, \sigma^2)$ independently, so that

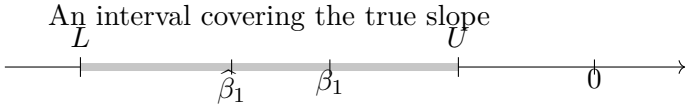
$$Y_i \sim N(\beta_0 + \beta_1 x_i, \sigma^2).$$

A point estimator gives a single estimate of an unknown parameter. An interval estimator gives an interval intended to cover the true parameter. We will consider confidence intervals for β_0, β_1, σ^2, and the mean response.

Definition 4.1 (Confidence interval)**.** For a fixed parameter θ and $0 < \alpha < 1$, an interval $[L(Y), U(Y)]$ is a $100(1-\alpha)\%$ confidence interval if

$$\mathbb{P}_\theta\{L(Y) \le \theta \le U(Y)\} = 1 - \alpha$$

for every parameter value under consideration. A conservative interval has coverage at least $1 - \alpha$. The probability refers to the random endpoints before the data are observed; the parameter itself is fixed.



Example 4.2 (One normal observation)**.** Let $n = 1$ and $X_1 \sim N(\mu, 1)$. We seek endpoints L and U such that

$$\mathbb{P}(\mu \in [L, U]) = \mathbb{P}(L \le \mu \le U) \simeq 0.95.$$

The event can be rewritten in the same order as

$$\begin{aligned}\{L \le \mu \le U\} &= \{L - X_1 \le \mu - X_1 \le U - X_1\} \\ &= \{X_1 - U \le X_1 - \mu \le X_1 - L\}.\end{aligned}$$

Since $X_1 - \mu \sim N(0, 1)$, choose

$$X_1 - U = -1.96, \qquad X_1 - L = 1.96.$$

It follows that

$$L = X_1 - 1.96, \qquad U = X_1 + 1.96.$$

Using $z_{0.975}$ in place of the rounded value 1.96 gives coverage exactly 0.95.

4.2 Confidence intervals for the regression parameters

For a distribution with a continuous distribution function, use lower-tail quantiles: z_q is the q quantile of $N(0, 1)$, $t_{\nu,q}$ is the q quantile of t_ν, and $\chi^2_{\nu,q}$ is the q quantile of χ^2_ν.

Exercise 4.1. Under the normal error model, obtain two-sided $100(1 - \alpha)\%$ confidence intervals for β_0, β_1, and σ^2.

Solution. The standard errors are

$$\mathrm{SE}(\widehat{\beta}_1) = \frac{s}{\sqrt{S_{xx}}}, \qquad \mathrm{SE}(\widehat{\beta}_0) = s\sqrt{\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}}}.$$

By the orthogonal-coordinate argument in Exercise 3.3, the coefficient vector $(\widehat{\beta}_0, \widehat{\beta}_1)$ is independent of s^2, and $(n - 2)s^2/\sigma^2 \sim \chi^2_{n-2}$. Consequently,

$$\frac{\widehat{\beta}_j - \beta_j}{\mathrm{SE}(\widehat{\beta}_j)} \sim t_{n-2}, \qquad j = 0, 1.$$

Starting from

$$\mathbb{P}\left\{-t_{n-2,1-\alpha/2} \le \frac{\widehat{\beta}_j - \beta_j}{\mathrm{SE}(\widehat{\beta}_j)} \le t_{n-2,1-\alpha/2}\right\} = 1 - \alpha$$

and rearranging gives the intervals

$$\begin{aligned}\beta_1 : &\quad \widehat{\beta}_1 \ \pm \ t_{n-2,1-\alpha/2}\frac{s}{\sqrt{S_{xx}}}, \\ \beta_0 : &\quad \widehat{\beta}_0 \ \pm \ t_{n-2,1-\alpha/2}\, s\sqrt{\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}}}.\end{aligned}$$

For the variance, use

$$\mathbb{P}\left\{\chi^2_{n-2,\alpha/2} \le \frac{(n-2)s^2}{\sigma^2} \le \chi^2_{n-2,1-\alpha/2}\right\} = 1 - \alpha.$$

All quantities are positive, so taking reciprocals reverses the inequalities. The interval is

$$\left[\frac{(n-2)s^2}{\chi^2_{n-2,1-\alpha/2}}, \quad \frac{(n-2)s^2}{\chi^2_{n-2,\alpha/2}}\right].$$

For the two-sided test of $H_0 : \beta_1 = 0$ at level α, rejection is equivalent to zero being outside the corresponding confidence interval for β_1. This follows directly from

$$\left|\frac{\widehat{\beta}_1}{s/\sqrt{S_{xx}}}\right| > t_{n-2,1-\alpha/2}.$$

4.3 Confidence interval on the mean response

Let x_0 be the level of the regressor variable for which we wish to estimate the mean response. With a new response at that fixed level,

$$\mu_0 = \mathbb{E}(Y_0 \mid x_0) = \mathbb{E}(\beta_0 + \beta_1 x_0 + \varepsilon_0) = \beta_0 + \beta_1 x_0.$$

Its point estimator is

$$\widehat{\mu}_0 = \widehat{\beta}_0 + \widehat{\beta}_1 x_0 = \bar{Y} + \widehat{\beta}_1(x_0 - \bar{x}).$$

Because $\widehat{\beta}_0$ and $\widehat{\beta}_1$ are unbiased,

$$\mathbb{E}(\widehat{\mu}_0) = \beta_0 + \beta_1 x_0 = \mu_0.$$

Under normal errors, $\widehat{\mu}_0$ follows a normal distribution because it is a linear combination of the Y_i.

Exercise 4.2. Find $\mathrm{Var}(\widehat{\mu}_0)$ by writing the estimator in terms of $\bar{Y}$ and $\widehat{\beta}_1$, and obtain a confidence interval for μ_0 under normal errors.

Solution. Recall that $\mathrm{Cov}(\bar{Y}, \widehat{\beta}_1) = 0$. Therefore,

$$\begin{aligned}
\mathrm{Var}(\widehat{\beta}_0 + \widehat{\beta}_1 x_0) &= \mathrm{Var}\{\bar{Y} + \widehat{\beta}_1(x_0 - \bar{x})\} \\
&= \mathrm{Var}(\bar{Y}) + (x_0 - \bar{x})^2 \,\mathrm{Var}(\widehat{\beta}_1) + 2(x_0 - \bar{x})\,\mathrm{Cov}(\bar{Y}, \widehat{\beta}_1) \\
&= \frac{\sigma^2}{n} + \frac{\sigma^2 (x_0 - \bar{x})^2}{S_{xx}} \\
&= \sigma^2 \left\{\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}}\right\}.
\end{aligned}$$

Define

$$h_0 = \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}}.$$

Then

$$\widehat{\mu}_0 \sim N(\mu_0, \sigma^2 h_0), \qquad \mathrm{SE}(\widehat{\mu}_0) = s\sqrt{h_0}.$$

Independence of the fitted estimators and the residual sum of squares gives the pivot

$$T = \frac{\widehat{\mu}_0 - \mu_0}{s\sqrt{h_0}} \sim t_{n-2}.$$

For $\alpha = 0.05$,

$$0.95 = \mathbb{P}\{-t_{n-2,0.975} \le T \le t_{n-2,0.975}\}.$$

In general, rearranging the central probability yields

$$\widehat{\beta}_0 + \widehat{\beta}_1 x_0 \;\pm\; t_{n-2,1-\alpha/2}\, s\sqrt{\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}}}. \tag{4.1}$$

For a fixed fitted data set, the interval width widens as $|x_0 - \bar{x}|$ increases. It is smallest at $x_0 = \bar{x}$, where the standard error is $s/\sqrt{n}$. Formula (4.1) gives a confidence interval at a specified x_0; its stated coverage is pointwise.

4.3.1 The convergence of the t distribution to normal

Exercise 4.3. Let $Z \sim N(0,1)$ and $V_\nu \sim \chi^2_\nu$ be independent. Show that

$$T_\nu = \frac{Z}{\sqrt{V_\nu/\nu}} \quad \text{converges in distribution to } N(0,1) \quad \text{as } \nu \longrightarrow \infty.$$

Solution. Write $V_\nu = \sum_{i=1}^{\nu} Z_i^2$, where the Z_i are independent standard normal variables. Since $\mathbb{E}(Z_i^2) = 1$, the law of large numbers gives

$$\frac{V_\nu}{\nu} = \frac{1}{\nu}\sum_{i=1}^{\nu} Z_i^2 \xrightarrow{\mathbb{P}} 1.$$

The square-root function is continuous at 1, so $\sqrt{V_\nu/\nu} \xrightarrow{\mathbb{P}} 1$. Slutsky's theorem now gives

$$T_\nu = \frac{Z}{\sqrt{V_\nu/\nu}} \xrightarrow{d} Z \sim N(0,1).$$

Thus the t_{n-2} distribution approaches the standard normal distribution as the sample size grows.

4.4 Prediction of a new observation

Given x_0, suppose we wish to predict a new observation

$$Y_0 = \beta_0 + \beta_1 x_0 + \varepsilon_0.$$

Assume ε_0 has mean zero and variance σ^2 and is independent of all errors used to fit the line. We can use

$$\widehat{Y}_0 = \widehat{\beta}_0 + \widehat{\beta}_1 x_0$$

as the predictor. Although it has the same numerical value as the estimator of the mean response, the prediction error also includes the new error ε_0.

Exercise 4.4. Compute the mean and variance of $\widehat{Y}_0 - Y_0$. Assuming independent normal errors, including the new error, obtain a prediction interval for Y_0 and compare it with the confidence interval for the mean response.

Solution. Unbiasedness and $\mathbb{E}(\varepsilon_0) = 0$ give

$$\mathbb{E}(\widehat{Y}_0 - Y_0) = 0.$$

Independence of the new observation and the fitted estimators gives

$$\operatorname{Var}(\widehat{Y}_0 - Y_0) = \operatorname{Var}(\widehat{\beta}_0) + x_0^2 \operatorname{Var}(\widehat{\beta}_1) + 2x_0 \operatorname{Cov}(\widehat{\beta}_0, \widehat{\beta}_1) + \operatorname{Var}(Y_0).$$

Since $\widehat{\beta}_0 = \bar{Y} - \bar{x}\widehat{\beta}_1$ and $\operatorname{Cov}(\bar{Y}, \widehat{\beta}_1) = 0$,

$$\operatorname{Cov}(\widehat{\beta}_0, \widehat{\beta}_1) = -\bar{x} \operatorname{Var}(\widehat{\beta}_1) = -\frac{\bar{x}\sigma^2}{S_{xx}}.$$

Substituting gives the computation

$$\begin{aligned} \operatorname{Var}(\widehat{Y}_0 - Y_0) &= \sigma^2 \left(\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}} \right) + \frac{x_0^2 \sigma^2}{S_{xx}} + \sigma^2 - \frac{2x_0 \bar{x} \sigma^2}{S_{xx}} \\ &= \sigma^2 \left\{ 1 + \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}} \right\} = \sigma^2 (1 + h_0). \end{aligned}$$

Equivalently, write $\widehat{Y}_0 - Y_0 = (\widehat{\mu}_0 - \mu_0) - \varepsilon_0$ and add the two independent variances.

Under the normal model the prediction error is normal. It is independent of s^2, because the fitted estimators are independent of the residual sum of squares and ε_0 is independent of all the original observations. Consequently,

$$\frac{\widehat{Y}_0 - Y_0}{s\sqrt{1 + h_0}} \sim t_{n-2}.$$

The prediction interval is

$$\widehat{\beta}_0 + \widehat{\beta}_1 x_0 \;\pm\; t_{n-2,1-\alpha/2}\, s \sqrt{1 + \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}}}. \tag{4.2}$$

The extra 1 inside the square root accounts for the variability due to ε_0. At the same x_0 and confidence level, the prediction interval is wider than the confidence interval for the mean response whenever $s > 0$; this occurs with probability one under the normal model with $\sigma^2 > 0$. The ratio of their half-widths is

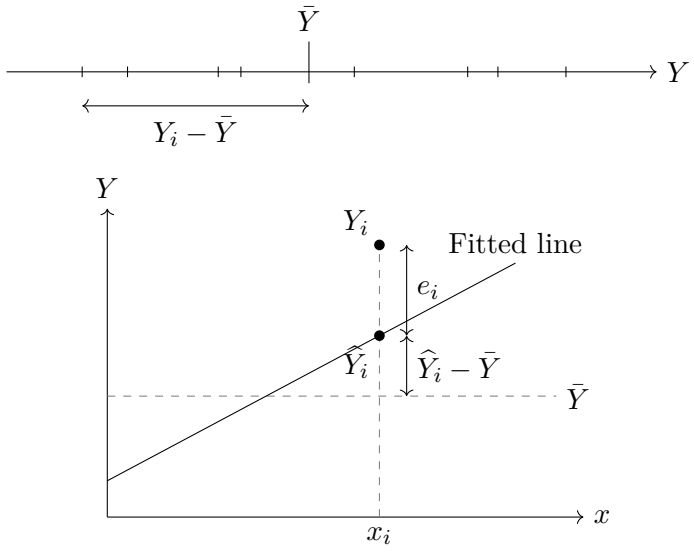
$$\sqrt{\frac{1 + h_0}{h_0}} > 1.$$

4.5 Analysis of variance

The variability of the responses around their sample mean is measured by the sample variance

$$\frac{1}{n-1} \sum_{i=1}^{n} (Y_i - \bar{Y})^2.$$

Regression separates this variability into a part represented by the fitted line and a residual part.



For each observation,

$$Y_i - \bar{Y} = (Y_i - \widehat{Y}_i) + (\widehat{Y}_i - \bar{Y}) = e_i + (\widehat{Y}_i - \bar{Y}). \tag{4.3}$$

The least-squares normal equations for a model with an intercept give

$$\sum_i e_i = 0, \qquad \sum_i x_i e_i = 0.$$

Hence fitted values are perpendicular to residuals in the sense that

$$\sum_i \widehat{Y}_i e_i = \widehat{\beta}_0 \sum_i e_i + \widehat{\beta}_1 \sum_i x_i e_i = 0.$$

Also,

$$\frac{1}{n}\sum_i \widehat{Y}_i = \frac{1}{n}\sum_i (Y_i - e_i) = \bar{Y}.$$

Exercise 4.5. Use these orthogonality identities to show that the total sum of squares is the sum of the regression and error sums of squares.

Solution. The cross-product in the square of (4.3) is zero:

$$\begin{aligned}\sum_i (\widehat{Y}_i - \bar{Y})(Y_i - \widehat{Y}_i) &= \sum_i (\widehat{Y}_i - \bar{Y}) e_i \\ &= \sum_i \widehat{Y}_i e_i - \sum_i \bar{Y} e_i \\ &= 0 - \bar{Y} \sum_i e_i = 0.\end{aligned}$$

Therefore,

$$\begin{aligned}\sum_i (Y_i - \bar{Y})^2 &= \sum_i \{(\widehat{Y}_i - \bar{Y}) + e_i\}^2 \\ &= \sum_i (\widehat{Y}_i - \bar{Y})^2 + \sum_i e_i^2.\end{aligned}$$

Define

$$\begin{aligned} \mathrm{SST} &= \sum_i (Y_i - \bar{Y})^2 && \text{(total sum of squares)}, \\ \mathrm{SSR} &= \sum_i (\widehat{Y}_i - \bar{Y})^2 && \text{(regression sum of squares)}, \\ \mathrm{SSE} &= \sum_i (Y_i - \widehat{Y}_i)^2 && \text{(error sum of squares)}. \end{aligned}$$

Then

$$\mathrm{SST} = \mathrm{SSR} + \mathrm{SSE}. \tag{4.4}$$

The centered response vector has $n-1$ degrees of freedom. Its fitted component is a multiple of $(x_i - \bar{x})_{i=1}^n$, which has dimension one. The residual vector satisfies the two independent constraints $\sum_i e_i = 0$ and $\sum_i x_i e_i = 0$, leaving dimension $n-2$. Thus the degrees of freedom also decompose:

$$n - 1 = 1 + (n - 2).$$

Source	Sum of squares	Degrees of freedom	Mean square
Regression	SSR	1	SSR
Error	SSE	$n-2$	SSE/$(n-2)$
Total	SST	$n-1$	

These are algebraic least-squares identities and do not require normal errors.

4.5.1 Coefficient of determination

For $\mathrm{SST} > 0$, the quantity

$$R^2 = \frac{\mathrm{SSR}}{\mathrm{SST}} = 1 - \frac{\mathrm{SSE}}{\mathrm{SST}} \tag{4.5}$$

is the coefficient of determination. It is the proportion of the total observed variation in the response represented by the fitted regression. Because all three sums of squares are nonnegative and satisfy (4.4), $0 \le R^2 \le 1$. If all responses are equal, $\mathrm{SST} = 0$ and this ratio is undefined.

The magnitude of R^2 depends on the range of variability in x. Indeed,

$$\widehat{Y}_i - \bar{Y} = \widehat{\beta}_1 (x_i - \bar{x}), \qquad \mathrm{SSR} = \widehat{\beta}_1^2 S_{xx}.$$

Thus both the fitted slope and the spread of the regressor values enter the explained sum of squares. Values of R^2 from data sets with different ranges of x need to be interpreted in that context.

4.6 Regression through the origin

Suppose instead that the model is constrained to pass through the origin:

$$Y_i = \beta_1 x_i + \varepsilon_i.$$

There is no intercept parameter. Assume fixed x_i, independent mean-zero errors of common variance σ^2, and $\sum_i x_i^2 > 0$. The least-squares function is

$$S(b) = \sum_{i=1}^{n} (Y_i - bx_i)^2.$$

Exercise 4.6. Minimize $S(b)$, determine the distribution of the resulting slope estimator under normal errors, and determine whether the residuals necessarily sum to zero.

Solution. Differentiating,

$$S'(b) = -2\sum_i x_i(Y_i - bx_i), \qquad S''(b) = 2\sum_i x_i^2 > 0.$$

The unique minimizer is

$$\widehat{\beta}_1 = \frac{\sum_i x_i Y_i}{\sum_i x_i^2}.$$

Substitute $Y_i = \beta_1 x_i + \varepsilon_i$ to obtain

$$\widehat{\beta}_1 = \beta_1 + \frac{\sum_i x_i \varepsilon_i}{\sum_i x_i^2}.$$

Hence

$$\mathbb{E}(\widehat{\beta}_1) = \beta_1, \qquad \operatorname{Var}(\widehat{\beta}_1) = \frac{\sigma^2}{\sum_i x_i^2}.$$

Under independent normal errors, it is a linear combination of normal variables, giving

$$\widehat{\beta}_1 \sim N\left(\beta_1, \frac{\sigma^2}{\sum_i x_i^2}\right).$$

The normal equation imposes $\sum_i x_i e_i = 0$, but the residual sum is

$$\sum_i e_i = \sum_i Y_i - \widehat{\beta}_1 \sum_i x_i = \sum_i Y_i - \frac{(\sum_i x_i Y_i)(\sum_i x_i)}{\sum_i x_i^2},$$

which need not vanish. For example, with

$$(x_1, x_2, x_3) = (1, 2, 3), \qquad (Y_1, Y_2, Y_3) = (1, 1, 1),$$

we get

$$\widehat{\beta}_1 = \frac{6}{14} = \frac{3}{7}, \qquad (e_1, e_2, e_3) = \left(\frac{4}{7}, \frac{1}{7}, -\frac{2}{7}\right), \qquad \sum_i e_i = \frac{3}{7} \neq 0, \quad \sum_i x_i e_i = 0.$$

The centered decomposition $\mathrm{SST} = \mathrm{SSR} + \mathrm{SSE}$ and the resulting interpretation of R^2 above rely on an intercept and cannot simply be carried over to a model through the origin. In the example,

$$\sum_i (Y_i - \bar{Y})^2 = 0, \quad \sum_i (\widehat{Y}_i - \bar{Y})^2 = \frac{21}{49}, \quad \sum_i e_i^2 = \frac{21}{49},$$

so the centered sum-of-squares identity fails.

Chapter 5

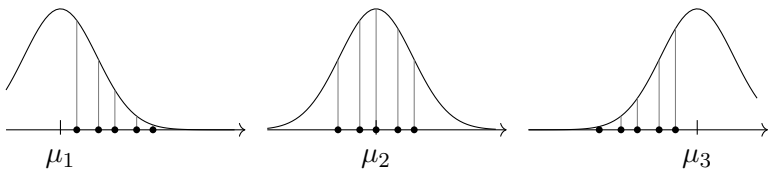
Maximum Likelihood Estimation

5.1 Maximum likelihood for a normal sample

Let $X_1, \dots, X_n$ be independent with a common distribution $N(\mu, \sigma^2)$. A density describes how observations are distributed for fixed parameter values. After observations $x_1, \dots, x_n$ are fixed, the same product of densities, viewed as a function of the parameters, is the likelihood function:

$$L(\mu, \sigma^2 \mid x_1, \dots, x_n) = \prod_{i=1}^{n} f(x_i; \mu, \sigma^2).$$

One can think of moving the normal distribution relative to the observed points and comparing the products of its density heights.



Maximizing L is equivalent to maximizing $\log L$, since the logarithm is strictly increasing.

Exercise 5.1. Write the normal likelihood and log-likelihood explicitly. Find the maximum likelihood estimators of μ and σ^2 when $\sum_i (x_i - \bar{x})^2 > 0$.

Solution. For $v = \sigma^2 > 0$,

$$L(\mu, v \mid x_1, \dots, x_n) = (2\pi v)^{-n/2} \exp\left\{ -\frac{1}{2v} \sum_i (x_i - \mu)^2 \right\},$$

$$\ell(\mu, v) = \log L(\mu, v) = -\frac{n}{2} \log(2\pi) - \frac{n}{2} \log v - \frac{1}{2v} \sum_i (x_i - \mu)^2.$$

For fixed v, the first two terms are constant in μ, and maximizing ℓ amounts to minimizing $\sum_i (x_i - \mu)^2$. The identity

$$\sum_i (x_i - \mu)^2 = \sum_i (x_i - \bar{x})^2 + n(\mu - \bar{x})^2$$

shows that the maximizing mean is $\widehat{\mu} = \bar{x}$. It also follows by differentiation:

$$\frac{\partial \ell}{\partial \mu} = \frac{1}{v}\sum_i (x_i - \mu) = 0 \quad \Longleftrightarrow \quad \mu = \bar{x}.$$

Let $Q = \sum_i (x_i - \bar{x})^2 > 0$. Substitution leaves the profile log-likelihood

$$\ell(\bar{x}, v) = -\frac{n}{2}\log(2\pi) - \frac{n}{2}\log v - \frac{Q}{2v},$$

whose derivative is

$$\frac{d}{dv}\ell(\bar{x}, v) = -\frac{n}{2v} + \frac{Q}{2v^2} = \frac{Q - nv}{2v^2}.$$

It is positive for $v < Q/n$ and negative for $v > Q/n$, proving that the unique global maximum occurs at

$$\widehat{\mu} = \bar{x}, \qquad \widehat{\sigma}^2_{\mathrm{ML}} = \frac{1}{n}\sum_i (x_i - \bar{x})^2.$$

If $Q = 0$, all observations coincide. Taking $\mu = \bar{x}$ and $v \downarrow 0$ makes the likelihood unbounded, so there is no finite maximum within the parameter space $v > 0$.

5.2 Likelihood and least squares

Maximum likelihood estimation chooses parameter values that maximize the likelihood of the observed data under a specified distribution. In simple linear regression, suppose for this section that

$$Y_i = \beta_0 + \beta_1 x_i + \varepsilon_i, \qquad \varepsilon_i \overset{\text{ind}}{\sim} N(0, \sigma^2), \qquad i = 1, \ldots, n,$$

where the x_i are fixed, $\sigma^2 > 0$, $n > 2$, and $S_{xx} = \sum_{i=1}^n (x_i - \bar{x})^2 > 0$. Independence gives a product of densities:

$$L(\beta_0, \beta_1, \sigma^2) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left\{-\frac{(y_i - \beta_0 - \beta_1 x_i)^2}{2\sigma^2}\right\},$$

$$\ell(\beta_0, \beta_1, \sigma^2) = \log L = -\frac{n}{2}\log(2\pi) - \frac{n}{2}\log\sigma^2 - \frac{1}{2\sigma^2}\sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i)^2.$$

The logarithm changes a product into a sum. Since it is increasing, maximizing L is equivalent to maximizing ℓ. For fixed σ^2, this is equivalent to minimizing the squared error sum.

Exercise 5.2. Differentiate the log likelihood with respect to β_0, β_1, and σ^2 to obtain the maximum likelihood estimates. Compare them with the least squares estimates.

Solution. The coefficient equations are

$$\frac{\partial \ell}{\partial \beta_0} = \frac{1}{\sigma^2}\sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i) = 0, \qquad \frac{\partial \ell}{\partial \beta_1} = \frac{1}{\sigma^2}\sum_{i=1}^n x_i(y_i - \beta_0 - \beta_1 x_i) = 0.$$

The first gives $\beta_0 = \bar{y} - \beta_1 \bar{x}$. Substituting into the second gives $S_{xy} - \beta_1 S_{xx} = 0$, where $S_{xy} = \sum_i (x_i - \bar{x})(y_i - \bar{y})$. Thus

$$\widetilde{\beta}_1 = \frac{S_{xy}}{S_{xx}} = \widehat{\beta}_1, \qquad \widetilde{\beta}_0 = \bar{y} - \widetilde{\beta}_1 \bar{x} = \widehat{\beta}_0.$$

Because $S_{xx} > 0$, the squared error sum has a unique minimum in the two coefficients. Let

$$\mathrm{SSE} = \sum_{i=1}^{n} (y_i - \widehat{\beta}_0 - \widehat{\beta}_1 x_i)^2.$$

Writing $v = \sigma^2$, the remaining derivative at these coefficients is

$$\frac{\partial \ell}{\partial v} = -\frac{n}{2v} + \frac{\mathrm{SSE}}{2v^2}.$$

For $\mathrm{SSE} > 0$, this derivative is positive when $v < \mathrm{SSE}/n$ and negative when $v > \mathrm{SSE}/n$. Therefore

$$\widetilde{\sigma}^2 = \frac{\mathrm{SSE}}{n}, \qquad \widehat{\sigma}^2 = \frac{\mathrm{SSE}}{n-2}, \qquad \widetilde{\sigma}^2 = \frac{n-2}{n}\widehat{\sigma}^2.$$

Here $\widehat{\sigma}^2$ denotes the usual unbiased variance estimator. The coefficient estimates agree with least squares, but the variance estimates use different divisors; the maximum likelihood variance estimate is smaller. If $\mathrm{SSE} = 0$, the likelihood is unbounded as $\sigma^2 \downarrow 0$, so there is no finite maximizer with $\sigma^2 > 0$. Under the stated normal model and $n > 2$, this exceptional case has probability zero.

Chapter 6

Matrix Formulation and Geometry of Linear Regression

6.1 Multiple linear regression in matrix form

A multiple linear regression model with $p-1$ regressors is

$$Y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_{p-1} x_{i,p-1} + \varepsilon_i, \qquad i = 1, \ldots, n. \tag{6.1}$$

Throughout this chapter, p is the total number of parameters, including the intercept. "Linear" means linear in the coefficients, not necessarily in the covariates. Matrix notation makes coefficient estimation easier.

For example, the first two observations satisfy

$$\begin{aligned} Y_1 &= \beta_0 + \beta_1 x_{11} + \cdots + \beta_{p-1} x_{1,p-1} + \varepsilon_1, \\ Y_2 &= \beta_0 + \beta_1 x_{21} + \cdots + \beta_{p-1} x_{2,p-1} + \varepsilon_2. \end{aligned}$$

Stacking all observations gives

$$\underbrace{\begin{pmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_n \end{pmatrix}}_{Y \in \mathbb{R}^n} = \underbrace{\begin{pmatrix} 1 & x_{11} & \cdots & x_{1,p-1} \\ 1 & x_{21} & \cdots & x_{2,p-1} \\ \vdots & \vdots & & \vdots \\ 1 & x_{n1} & \cdots & x_{n,p-1} \end{pmatrix}}_{X \in \mathbb{R}^{n \times p}} \underbrace{\begin{pmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_{p-1} \end{pmatrix}}_{\beta \in \mathbb{R}^p} + \underbrace{\begin{pmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \vdots \\ \varepsilon_n \end{pmatrix}}_{\varepsilon \in \mathbb{R}^n}. \tag{6.2}$$

Thus $Y = X\beta + \varepsilon$. The design matrix X is fixed. Assume that the errors are independent with

$$\mathbb{E}(\varepsilon_i) = 0, \qquad \mathrm{Var}(\varepsilon_i) = \sigma^2 > 0.$$

For the inverse formulas and variance estimation below, assume $\mathrm{rank}(X) = p$ and $n > p$. Normality is not required for the least squares and covariance results that follow.

6.1.1 The estimation space

Let $\mathbf{1} = (1, \ldots, 1)^T$ and $x_j = (x_{1j}, \ldots, x_{nj})^T$. Then

$$X = \begin{pmatrix} \mathbf{1} & x_1 & \cdots & x_{p-1} \end{pmatrix}, \qquad X\beta = \beta_0 \mathbf{1} + \beta_1 x_1 + \cdots + \beta_{p-1} x_{p-1}.$$

Thus the mean vector is a linear combination of the columns of X. The vector of fitted values has the same form, with estimated coefficients:

$$\widehat{y} = X\widehat{\beta} = \widehat{\beta}_0\mathbf{1} + \widehat{\beta}_1 x_1 + \cdots + \widehat{\beta}_{p-1}x_{p-1}.$$

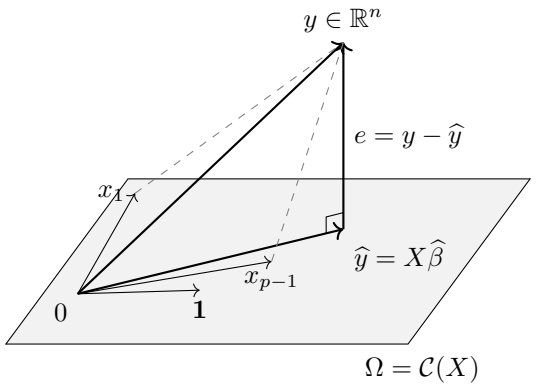
The observed response y also contains the error vector and need not be a linear combination of these columns.

Definition 6.1. The column space, or estimation space, is

$$\Omega = \mathcal{C}(X) = \{Xb : b \in \mathbb{R}^p\} = \left\{ b_0\mathbf{1} + \sum_{j=1}^{p-1} b_j x_j : b_0, \ldots, b_{p-1} \in \mathbb{R} \right\}.$$

It is a subspace of the n-dimensional observation space $\mathbb{R}^n$, with dimension $\operatorname{rank}(X) \le p$. Under full column rank, its dimension is p.

In particular, the column of ones is inside X, so $\mathbf{1} \in \mathcal{C}(X)$. Least squares is a projection onto this space, represented schematically by a plane below.



6.2 Least squares and orthogonal projection

The least squares criterion for an observed vector y is

$$Q(b) = \|y - Xb\|^2 = (y - Xb)^T(y - Xb).$$

Exercise 6.1. Derive the normal equations and the least squares estimator. Show that the fitted vector is the closest vector to y in $\mathcal{C}(X)$.

Solution. Expanding and differentiating gives

$$Q(b) = y^Ty - 2b^TX^Ty + b^TX^TXb, \qquad \nabla_b Q(b) = -2X^Ty + 2X^TXb.$$

The normal equations are

$$X^TX\widehat{\beta} = X^Ty.$$

Full column rank implies that X^TX is positive definite, since $a^TX^TXa = \|Xa\|^2 > 0$ for $a \neq 0$. Hence

$$\widehat{\beta} = (X^TX)^{-1}X^Ty, \qquad \widehat{y} = X\widehat{\beta} = Hy, \qquad H = X(X^TX)^{-1}X^T. \tag{6.3}$$

The normal equations also give

$$X^T e = X^T(y - X\widehat{\beta}) = 0.$$

Therefore the residual e is perpendicular to every column of X, and thus to all of $\mathcal{C}(X)$. For any $b \in \mathbb{R}^p$,

$$y - Xb = e + X(\widehat{\beta} - b),$$

where the two terms are orthogonal. Consequently,

$$\|y - Xb\|^2 = \|e\|^2 + \|X(\widehat{\beta} - b)\|^2 \geq \|e\|^2.$$

Equality holds only when $b = \widehat{\beta}$, by full column rank. This proves both minimization and the projection interpretation.

For simple linear regression, the normal equations take the two-parameter form:

$$\begin{pmatrix} n & \sum_i x_i \\ \sum_i x_i & \sum_i x_i^2 \end{pmatrix} \begin{pmatrix} \widehat{\beta}_0 \\ \widehat{\beta}_1 \end{pmatrix} = \begin{pmatrix} \sum_i y_i \\ \sum_i x_i y_i \end{pmatrix}.$$

The coefficient matrix has determinant nS_{xx} and is invertible under the same assumption. This is the two-parameter case of the matrix normal equations.

Exercise 6.2. Show that H and $I_n - H$ are symmetric and idempotent. Identify their projection spaces and write the residuals in matrix notation.

Solution. Since $X^T X$ and its inverse are symmetric,

$$H^T = \left[X(X^T X)^{-1} X^T\right]^T = H.$$

Direct multiplication gives

$$H^2 = X(X^T X)^{-1} X^T X (X^T X)^{-1} X^T = X(X^T X)^{-1} X^T = H.$$

Thus H is idempotent. Also $(I_n - H)^T = I_n - H$, and

$$\begin{aligned}(I_n - H)(I_n - H) &= I_n I_n - I_n H - H I_n + HH \\ &= I_n - H - H + H = I_n - H.\end{aligned}$$

For every v, $Hv \in \mathcal{C}(X)$; for every $Xb \in \mathcal{C}(X)$,

$$H(Xb) = X(X^T X)^{-1} X^T Xb = Xb.$$

Furthermore,

$$X^T(I_n - H) = 0, \qquad H(I_n - H) = (I_n - H)H = 0.$$

Hence H is the orthogonal projection onto $\mathcal{C}(X)$, and $I_n - H$ is the orthogonal projection onto its perpendicular complement. The difference between y_i and $\widehat{y}_i$ is the residual $e_i = y_i - \widehat{y}_i$. All n residuals are written as

$$e = y - \widehat{y} = y - Hy = (I_n - H)y. \tag{6.4}$$

In particular, $\widehat{y}^T e = 0$ and $\|y\|^2 = \|\widehat{y}\|^2 + \|e\|^2$. Since $\mathbf{1}$ is a column of X, $\mathbf{1}^T e = 0$: the residuals sum to zero.

6.3 Expectation and covariance of random vectors

Let $U = (U_1, \dots, U_m)^T$ and $V = (V_1, \dots, V_r)^T$ have finite second moments. Their cross-covariance matrix is

$$\begin{aligned}\operatorname{Cov}(U, V) &= \mathbb{E}\big[(U - \mathbb{E}U)(V - \mathbb{E}V)^T\big] \\ &= \begin{pmatrix} \operatorname{Cov}(U_1, V_1) & \cdots & \operatorname{Cov}(U_1, V_r) \\ \operatorname{Cov}(U_2, V_1) & \cdots & \operatorname{Cov}(U_2, V_r) \\ \vdots & & \vdots \\ \operatorname{Cov}(U_m, V_1) & \cdots & \operatorname{Cov}(U_m, V_r) \end{pmatrix}.\end{aligned}$$

The variance matrix of a random vector is its covariance with itself:

$$\begin{aligned}\operatorname{Var}(Y) = \operatorname{Cov}(Y, Y) &= \mathbb{E}\big[(Y - \mathbb{E}Y)(Y - \mathbb{E}Y)^T\big] \\ &= \begin{pmatrix} \operatorname{Var}(Y_1) & \cdots & \operatorname{Cov}(Y_1, Y_n) \\ \vdots & \ddots & \vdots \\ \operatorname{Cov}(Y_n, Y_1) & \cdots & \operatorname{Var}(Y_n) \end{pmatrix}.\end{aligned}$$

The centered column has size $n \times 1$ and its transpose has size $1 \times n$, so their product is an $n \times n$ matrix.

6.3.1 Linear transformations

For a constant matrix A and constant vector b of compatible sizes, linearity of expectation gives

$$\mathbb{E}(AY + b) = A\mathbb{E}(Y) + b.$$

The variance calculation proceeds by centering:

$$\begin{aligned}\operatorname{Var}(AY + b) &= \operatorname{Var}(AY) \\ &= \mathbb{E}\big[(AY - \mathbb{E}(AY))(AY - \mathbb{E}(AY))^T\big] \\ &= \mathbb{E}\big[A(Y - \mathbb{E}Y)\{A(Y - \mathbb{E}Y)\}^T\big] \\ &= A\mathbb{E}\big[(Y - \mathbb{E}Y)(Y - \mathbb{E}Y)^T\big]A^T \\ &= A \operatorname{Var}(Y) A^T.\end{aligned}$$

For a vector, the product $(Y - \mathbb{E}Y)(Y - \mathbb{E}Y)^T$ is an outer product; it is not a componentwise square.

Exercise 6.3. For constant matrices $A \in \mathbb{R}^{a \times m}$ and $B \in \mathbb{R}^{b \times r}$, find $\operatorname{Cov}(AU, BV)$.

Solution. Using the definition and the fact that A and B are constant,

$$\begin{aligned}\operatorname{Cov}(AU, BV) &= \mathbb{E}\big[(AU - A\mathbb{E}U)(BV - B\mathbb{E}V)^T\big] \\ &= \mathbb{E}\big[A(U - \mathbb{E}U)(V - \mathbb{E}V)^T B^T\big] \\ &= A \operatorname{Cov}(U, V) B^T.\end{aligned}$$

The resulting matrix has size $a \times b$. Adding fixed vectors to AU and BV does not change this covariance.

Exercise 6.4. Is $\mathrm{Cov}(U,V) = \mathrm{Cov}(V,U)$ for random vectors?

Solution. Scalar covariance is symmetric, but reversing two random vectors transposes their covariance matrix:

$$\begin{aligned}\mathrm{Cov}(V,U) &= \mathbb{E}\big[(V-\mathbb{E}V)(U-\mathbb{E}U)^T\big] \\ &= \Big\{\mathbb{E}\big[(U-\mathbb{E}U)(V-\mathbb{E}V)^T\big]\Big\}^T \\ &= \mathrm{Cov}(U,V)^T.\end{aligned}$$

When $m \neq r$, the two matrices even have different sizes. When their sizes agree, equality still need not hold. For instance, let Z have mean zero and variance one, and set

$$U = \begin{pmatrix} Z \\ 0 \end{pmatrix}, \qquad V = \begin{pmatrix} 0 \\ Z \end{pmatrix}.$$

Then

$$\mathrm{Cov}(U,V) = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \qquad \mathrm{Cov}(V,U) = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}.$$

Taking $U = V$ shows that a variance matrix is always symmetric.

6.4 Properties of the least squares estimator

The error assumptions give

$$\mathbb{E}(\varepsilon) = \begin{pmatrix} \mathbb{E}(\varepsilon_1) \\ \vdots \\ \mathbb{E}(\varepsilon_n) \end{pmatrix} = \begin{pmatrix} 0 \\ \vdots \\ 0 \end{pmatrix}.$$

Since distinct errors are independent, their covariances are zero. Consequently,

$$\begin{aligned}\mathrm{Var}(\varepsilon) &= \mathrm{Cov}(\varepsilon,\varepsilon) \\ &= \begin{pmatrix} \mathrm{Var}(\varepsilon_1) & \cdots & \mathrm{Cov}(\varepsilon_1,\varepsilon_n) \\ \vdots & \ddots & \vdots \\ \mathrm{Cov}(\varepsilon_n,\varepsilon_1) & \cdots & \mathrm{Var}(\varepsilon_n) \end{pmatrix} \\ &= \begin{pmatrix} \sigma^2 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \sigma^2 \end{pmatrix} = \sigma^2 I_n.\end{aligned}$$

Thus ε has mean zero and variance matrix $\sigma^2 I_n$. Since X and β are fixed,

$$\begin{aligned}\mathbb{E}(Y) &= \mathbb{E}(X\beta+\varepsilon) = \mathbb{E}(X\beta) + \mathbb{E}(\varepsilon) = X\beta, \\ \mathrm{Var}(Y) &= \mathrm{Var}(\varepsilon) = \sigma^2 I_n.\end{aligned}$$

In particular, for constant A, b,

$$\mathbb{E}(AY+b) = AX\beta + b, \qquad \mathrm{Var}(AY+b) = \sigma^2 AA^T.$$

Exercise 6.5. Find the expectation and variance matrix of $\widehat{\beta}$.

Solution. Let $C = (X^TX)^{-1}X^T$, so that $CX = I_p$. Viewing the estimator as the random vector $\widehat{\beta} = CY$,

$$\mathbb{E}(\widehat{\beta}) = C\mathbb{E}(Y) = CX\beta = \beta.$$

Thus the least squares estimator is unbiased. Its variance is

$$\begin{aligned}\operatorname{Var}(\widehat{\beta}) &= C\operatorname{Var}(Y)C^T = \sigma^2 CC^T \\ &= \sigma^2(X^TX)^{-1}X^TX(X^TX)^{-1} = \sigma^2(X^TX)^{-1}.\end{aligned}$$

The diagonal entries give individual coefficient variances, and the off-diagonal entries give covariances between coefficient estimates.

Exercise 6.6. Find the means and covariance matrices of the fitted vector $\widehat{Y} = HY$ and residual vector $e = (I_n - H)Y$. Compute their cross-covariance, and the cross-covariance of $\widehat{\beta}$ and e.

Solution. Because $HX = X$ and $(I_n - H)X = 0$,

$$\mathbb{E}(\widehat{Y}) = HX\beta = X\beta, \qquad \mathbb{E}(e) = (I_n - H)X\beta = 0.$$

Symmetry and idempotence give

$$\begin{aligned}\operatorname{Var}(\widehat{Y}) &= H(\sigma^2 I_n)H^T = \sigma^2 H, \\ \operatorname{Var}(e) &= (I_n - H)(\sigma^2 I_n)(I_n - H)^T = \sigma^2(I_n - H), \\ \operatorname{Cov}(\widehat{Y}, e) &= H(\sigma^2 I_n)(I_n - H)^T = 0.\end{aligned}$$

Moreover, with $C = (X^TX)^{-1}X^T$,

$$\operatorname{Cov}(\widehat{\beta}, e) = \sigma^2 C(I_n - H) = 0,$$

because $X^T(I_n - H) = 0$. These zero covariances establish uncorrelatedness. If the errors are additionally jointly normal, the corresponding pairs of vectors are jointly normal and therefore independent.

Writing $H = (h_{ij})$ makes the individual entries explicit:

$$\operatorname{Var}(\widehat{Y}_i) = \sigma^2 h_{ii}, \qquad \operatorname{Var}(e_i) = \sigma^2(1 - h_{ii}), \qquad \operatorname{Cov}(e_i, e_j) = -\sigma^2 h_{ij} \quad (i \neq j).$$

Independent errors therefore need not produce independent residuals.

Exercise 6.7. Determine the dimension of the residual space and use it to obtain an unbiased estimator of σ^2.

Solution. The image of H is $\mathcal{C}(X)$, which has dimension p. Its orthogonal complement, the image of $I_n - H$, has dimension $n - p$. A symmetric idempotent matrix has only eigenvalues zero and one: if $Hv = \lambda v$ with $v \neq 0$, then $H^2v = Hv$ gives $\lambda^2 = \lambda$. Thus its trace equals its rank, and

$$\operatorname{tr}(H) = p, \qquad \operatorname{tr}(I_n - H) = n - p.$$

For the random residual vector $e = (I_n - H)Y$, the residual sum of squares is

$$\mathrm{SSE} = e^Te = Y^T(I_n - H)Y.$$

Its observed value is $y^T(I_n - H)y$. Since $\mathbb{E}(e_i) = 0$,

$$\begin{aligned}\mathbb{E}(\text{SSE}) &= \sum_{i=1}^{n} \mathbb{E}(e_i^2) = \sum_{i=1}^{n} \text{Var}(e_i) \\ &= \sigma^2 \sum_{i=1}^{n} (1 - h_{ii}) = \sigma^2(n - p).\end{aligned}$$

Therefore

$$\widehat{\sigma}^2 = \frac{\text{SSE}}{n - p}$$

is unbiased for σ^2. With independent normal errors and $\text{SSE} > 0$, the likelihood calculation instead gives $\widetilde{\sigma}^2 = \text{SSE}/n$. In simple linear regression, $p = 2$, recovering the two divisors in Section 5.2.